MECH MAINT- TURBINE Department
SIMHADRI

Date:
21/12/2022(18:44:49)

Ref:SMTPP/MECH MAINT-/2022-23/LMI/10/698841

Subject: Approval for the LMI for Guidelines for ACV/MCV changeover in TDBFPs of BHEL 500MW units

LMI for "Guidelines for ACV/MCV changeover in TDBFPs of BHEL 500MW units"

LMI No: LMI/OGN/OPS/MECH/010 has been prepared in accordance with Operation guidance note COS-ISO-00-OGN/OPS/MECH/010 Rev. No.: 0 Sept 2022

Put up for Approval of Competent authority.

Submitted by:

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LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/010/

ISSUE NO.0; REVISION NO. 0; DATE: OCT'2022

GUIDELINES FOR ACV & MCV CHANGEOVER IN BHEL MAKE TDBFP DRIVE TURBINES

APPROVED FOR

Implementation---

***HOP*, SIMHADRI**

Date--------------------------

LMI APPROVAL DOCUMENTS

1.0 TITLE OF LMI	**GUIDELINES FOR ACV & MCV CHANGEOVER IN BHEL MAKE TDBFP DRIVE TURBINES**
2.a) DOCUMENT CODE **_b) REVISION/ISSUE NO.&DATE**	**LMI/OGN/OPS/MECH/010/** **ISSUE-0; REV.-0; OCT'2022**
3.ORIGINAL/REF.DOCUMENT	**COS-ISO-00-OGN/OPS/MECH/010 Rev. No.: 0 Sept 2022**
4.PRERPARED BY	**SENIOR MANAGER (TMD)**
5. CHECKED BY	**DGM (TMD)**
6.RECHECKED BY	**HOD(TMD)**
7.RECHECKED BY	**GM(MAINT)**
8. REVIEWED BY	**GM(O&M)**
9. APPROVED BY	**HOP**
10.INDEX LIST	**ENCLOSED**
11.DISTRIBUTION LIST	**ENCLOSED**

INDEX

2.0 INTRODUCTION

There are two nos. of Turbine driven BFPs & one/two nos. of Motor driven BFP in 500 MW, BHEL designed units. The Turbine driven BFP's are designed to cater for regular feed water supply while MDBFP is kept as stand by in unit operation. The sources of steam supply for Turbine driven BFP are either from IP exhaust, CRH or Aux PRDS. CRH & Aux PRDS steam source is provided as secondary source of steam during load operation or start up. The steam supply to drive turbine of TDBFP's controlled by Main Control valve (MCV) of TDBFP turbine from IP exhaust as per feed pump discharge pressure & flow via three element feed water controllers. Whenever there is sudden change in unit load and TDBFP is not able to meet feed water demand even after full opening of MCV, a command will go for opening of ACV & MCV will get throttled as per governing characteristic and maintain required steam flow to meet feed water demand as per speed/feed water controller.

At NTPC Simhadri it was observed that CRH block valve of steam supply to drive turbine of TDBFP is kept closed due to various problem of ACV i.e. hunting of ACV, valve sticking problem, over speeding of Drive turbine & tripping of drive turbine on inlet steam pressure high during MCV /ACV changeover causing tripping of TDBFP and unit.

3.0 SCOPE

The scope of this LMI is to provide guidelines & maintenance schedules to be followed for ensuring smooth changeover ACV and MCV in all BHEL Make TDBFP Drive Turbines.

4.0 REFERENCE DOCUMENT

Operation guidance note COS-ISO-00-OGN/OPS/MECH/010 Rev. No.: 0 Sept 2022

5.0 SUPERSEDED DOCUMENT

NIL

6.0 BRIEF OF THE PROBLEM

The governing characteristic of BFP Drive Turbine is designed such that whenever a demand of secondary source is required, a command goes to open Aux. Control Valve & Main Control Valve (MCV) is throttled to a preset value to control steam flow, speed of turbine and feed water flow of BFP as per the speed control /feed control demand. It has been observed that in case changeover of ACV/MCV is not

maintained.

Following are the various problems observed at various NTPC stations:

- **MCV /ACV problems: Turbine governing response is sluggish**
- **There is a command & feedback mismatch of MCV / ACV & ACV is not closing fully causing safety valve lifting whenever CRH block valve is kept opened.**
- **Tripping of drive turbine during MCV/ACV changeover due to turbine inlet steam pressure (Live Steam Pressure) high.**
- **Tripping of drive turbine on over speed during MCV/ACV changeover.**
- **Hunting of MCV & ACV**
- **Oil leakage from MCV/ ACV**
- **Drifting in HP & LP secondary pressure causing change in valve opening.**
- **Hunting & hammering in steam line during opening of ACV**
- **Recirculation valve passing causing increased steam flow in drive turbine leading to ACV opening.**

7.0 OBJECTIVE OF THE DOCUMENT

To increase reliability of TDBFP and avoid forced outage on drum level Low-Low during load variation & part load operation, safe & reliable operation of MCV & ACV changeover is required. Proper setting of MCV / ACV, start of opening & trimming is most important for smooth changeover of MCV/ACV and proper control of feed water as per feed controller output. Improper oil quality, moisture & mechanical impurities adversely affect MCV/ACV operation & I/H converter.

The Location Management Instruction has been made with the following objective in mind

1) **MCV /ACV setting i.e. Valve closure, start of opening & valve full travel to be set as per governing characteristic (in ANNEXURE).**
2) **To maintain oil quality, moisture & mechanical impurities within allowable limit**
3) **Cleaning, inspection & repair of servomotor for smooth operation of MCV & ACV**
4) **To ensure draining of moisture/condensate in CRH and Auxiliary Steam line to ACV to avoid moisture ingress in ACV.**

8.0 OPERATION AND MAINTENANCE PRACTICES TO BE FOLLOWED

S.No.	Issues	Action	Frequency	Responsibility
01	Disturbance in MCV/ACV Setting	MCV & ACV to be set as per governing characteristic (Ref Annexure-I for 500MW Units. For others, OEM recommendations to be followed): i) Full closure of MCV /ACV to be set and marked zero position on scale at 1.5 Kg/cm2 HP/LP secondary oil pressure. ii) Full travel of MCV/ACV to be set & marked on scale at 4.5 Kg/cm2 HP/LP secondary oil pressure. iii) Start opening of MCV/ACV to be set at 1.6 Kg/cm2 HP/LP secondary oil pressure and valve lift of 2 mm to be set. iv) Trimming of MCV/ACV changeover to be done as per governing characteristic, however fine tuning may be done as per pump requirement. v) Control oil pressure to be maintained within +/- 1.0 KSC during running w.r.t. pressure set during Governing setting. vi) Providing speed limiter in TDBFPs to prevent the pump going to runout and over speeding condition.	Setting to done after governing system overhaul of Drive turbine. Governing characteristic & MCV/ACV valve full closure, full travel to be jointly checked by Operation & TMD in every annual overhaul / Opportunity	Operation, TMD and C&I
02	Hunting in MCV / ACV	1. Maintaining of oil level in MOT as per drawing. Calibration of MOT level with ALONG C&I and Operation during every unit overhaul and maintaining joint protocol. 2. Cleaning of MCV/ACV pilot valves 3. Cleaning/ replacement of governing oil filter 4. Maintain Oil quality i.e. Moisture & Mechanical impurities with in allowable limit. External Filtration machine with multistage filtration with filter size of minimum 5 Micron to be taken in service whenever required to meet the oil parameters. 5. Servicing of servomotor if required 6. Replacement of bush provided in the top cover of pilot valve adjusting screw of MCV/ACV. 7. Accumulators Nitrogen pressure checking & correction once in a year as per COS guidelines. 8. Servicing of Safety relief valves provided at individual AOP discharge and PRV (pressure regulation valve) provided at pump discharge header	On opportunity shutdown/regular basis / Unit Overhaul	Operation, TMD and C&I
03	High moisture in lubrication oil	- Operation of centrifuge with heater in purifier mode - Attend seal water leakage from Main pump & booster pump - Ensure proper functioning of seal steam controller, set point & draining of moisture from seal steam header	As per requirement	Operation, TMD and C&I

04	CRH safety valve lifting	Drifting in HP/LP secondary pressure may lead to increase in HP/LP secondary pressure and MCV/ACV valve opening causes increase in steam inlet pressure. • Check & control moisture & mechanical impurities in oil • I/H converter mal function, replacement of I/H converter • Check & set MCV/ACV governing characteristic	Continuous basis / as per requirement	Chemistry, C&I, OPERATION and TMD
05	Servicing of servomotor of MCV /ACV	- Cleaning of complete servomotor along with pilot valve of MCV & ACV to be done during unit overhaul. - Servicing of ACV & MCV - Complete replacement of servomotor along with pilot valve of MCV & ACV.	Unit overhaul Unit overhaul Every four Years	TMD
06	Inspection of CRH/Aux Steam line drain	- Availability & thoroughness of drain in CRH & Aux Steam line to ACV is to be ensured. (Ref Annexure-2 for 500MW Units) - Availability & thoroughness of continuous orifice / drain trap in drain of CRH & Aux steam line to ACV (Ref Annexure-3 for 500MW Units)	Unit overhaul	TMD
07	Installation of continuous orifice in CRH drain line	Continuous orifice/drain trap in CRH/Aux steam drain line to be installed in old fleet of TDBFP or wherever it is not provided.	Opportunity shutdown/ Unit overhaul	TMD
08	TDBFP MCV/ACV changeover checking	Routine checking of MCV/ACV changeover on load to be carried out. SOP for above checking to be made and followed.	Quarterly	Operation
09	Platform near ACV & MCV of TDBFP	A platform near ACV & MCV to be made for proper access for ACV valve setting & checking	As per requirement	TMD
10	TDBFP recirculation valve passing	TDBFP recirculation valves passing defect is to be attended on priority as it unnecessarily loads the TDBFP, and pump operates at higher speed than design.	Opportunity shut down	TMD

9.0 RECORDS TO BE MAINTAINED

Sl No.	Record Description	Maintained by	Frequency	Remarks
1	Records must be kept for all activities carried out as mentioned above.	Responsibility of carrying out different activities shall be with Operation, TMD & C&I Department as indicated above.	As mentioned, Above.	The defects noticed during checking shall be attended during any available opportunity.

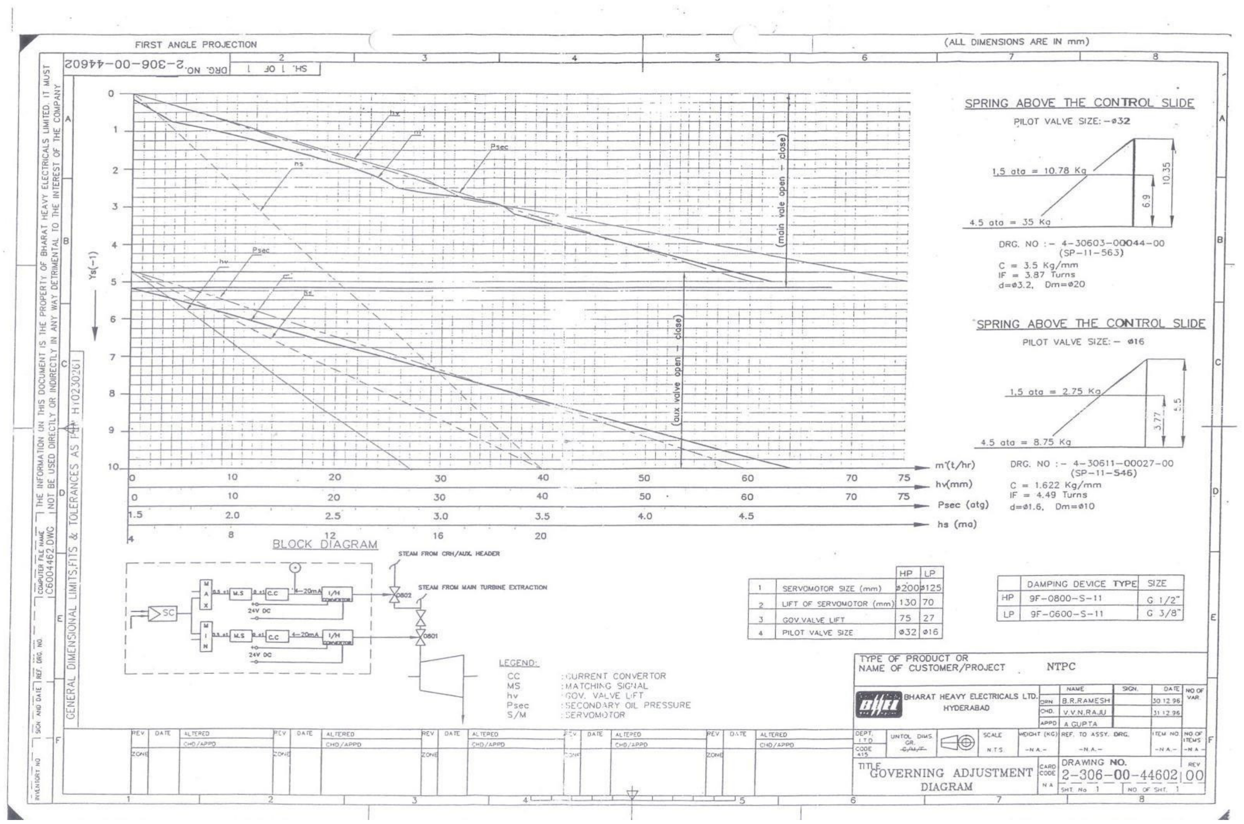
10.0 REVIEW

This document shall be reviewed by HOP, SIMHADRI on two-yearly basis.

11.0 ASSOCIATED DOCUMENTS

ANNEXURE-1: Governing Adjustment Diagram for 500MW BHEL Make units
ANNEXURE-2: Drain scheme of steam lines to TDBFP Drive Turbine (for Ref 500MW)
ANNEXURE-3: Details of Continuous drains / steam traps provided in drain scheme (for Ref 500MW)

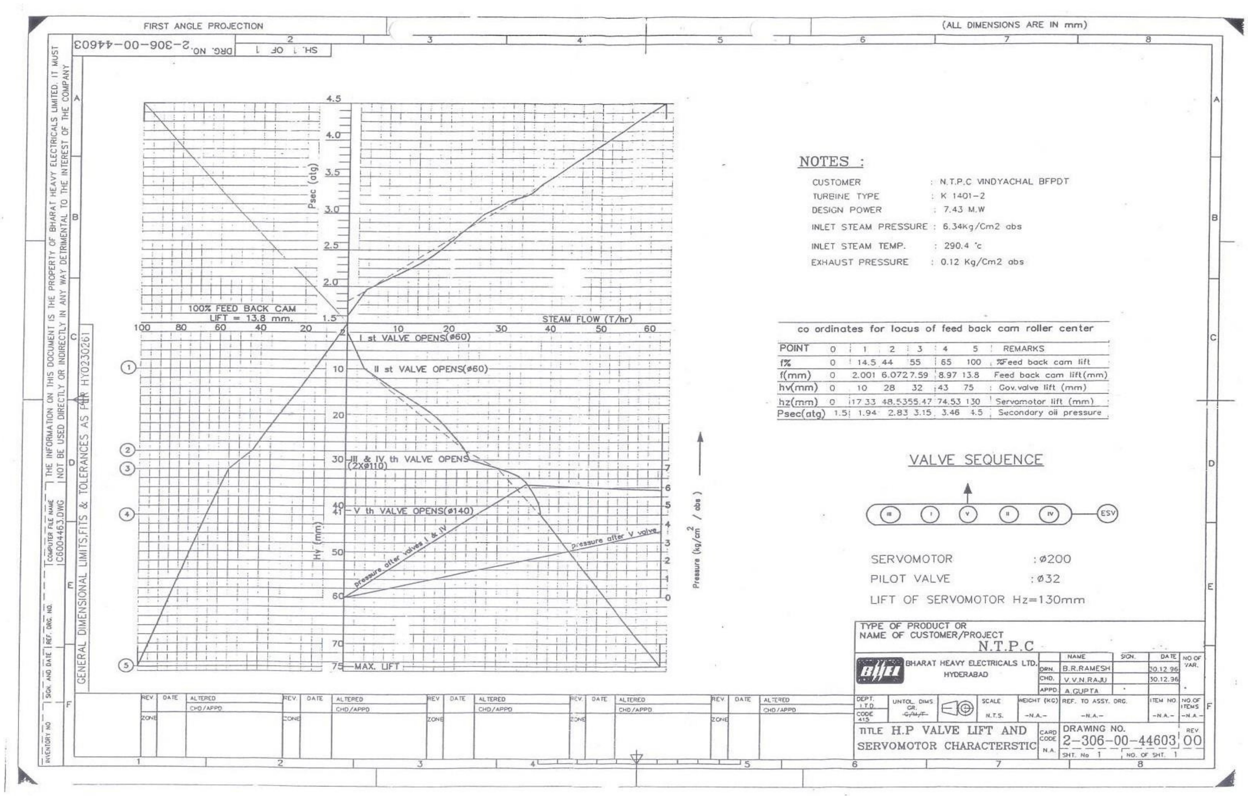
ANNEXURE-1: Governing Adjustment Diagram for 500MW BHEL Make units



		HP	LP
1	SERVOMOTOR SIZE (mm)	ø200	ø125
2	LIFT OF SERVOMOTOR (mm)	130	70
3	GOV.VALVE LIFT	75	27
4	PILOT VALVE SIZE	ø32	ø16

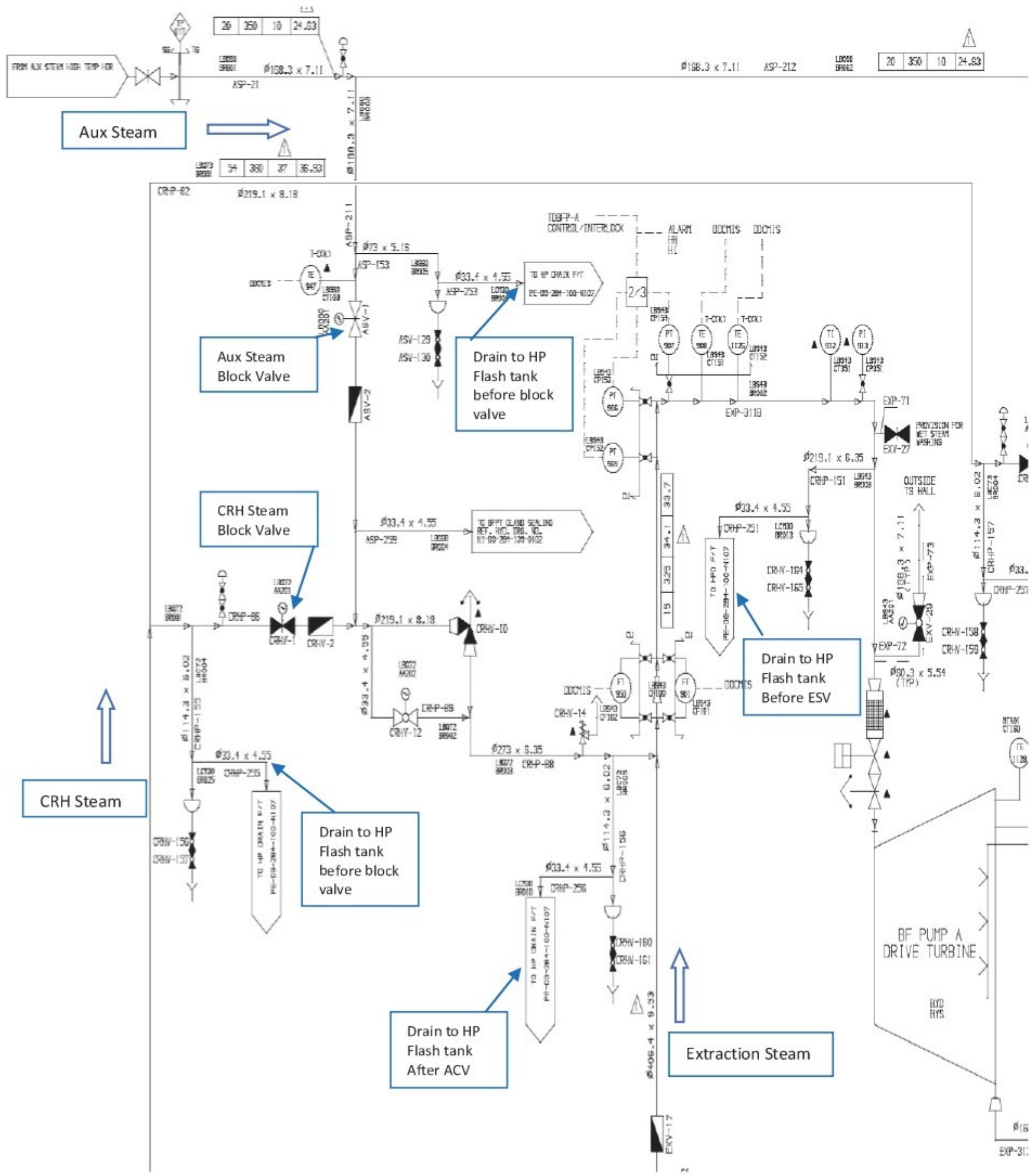
	DAMPING DEVICE TYPE	SIZE
HP	9F-0800-S-11	G 1/2"
LP	9F-0600-S-11	G 3/8"

ANNEXURE-1: Governing Adjustment Diagram for 500MW BHEL Make units

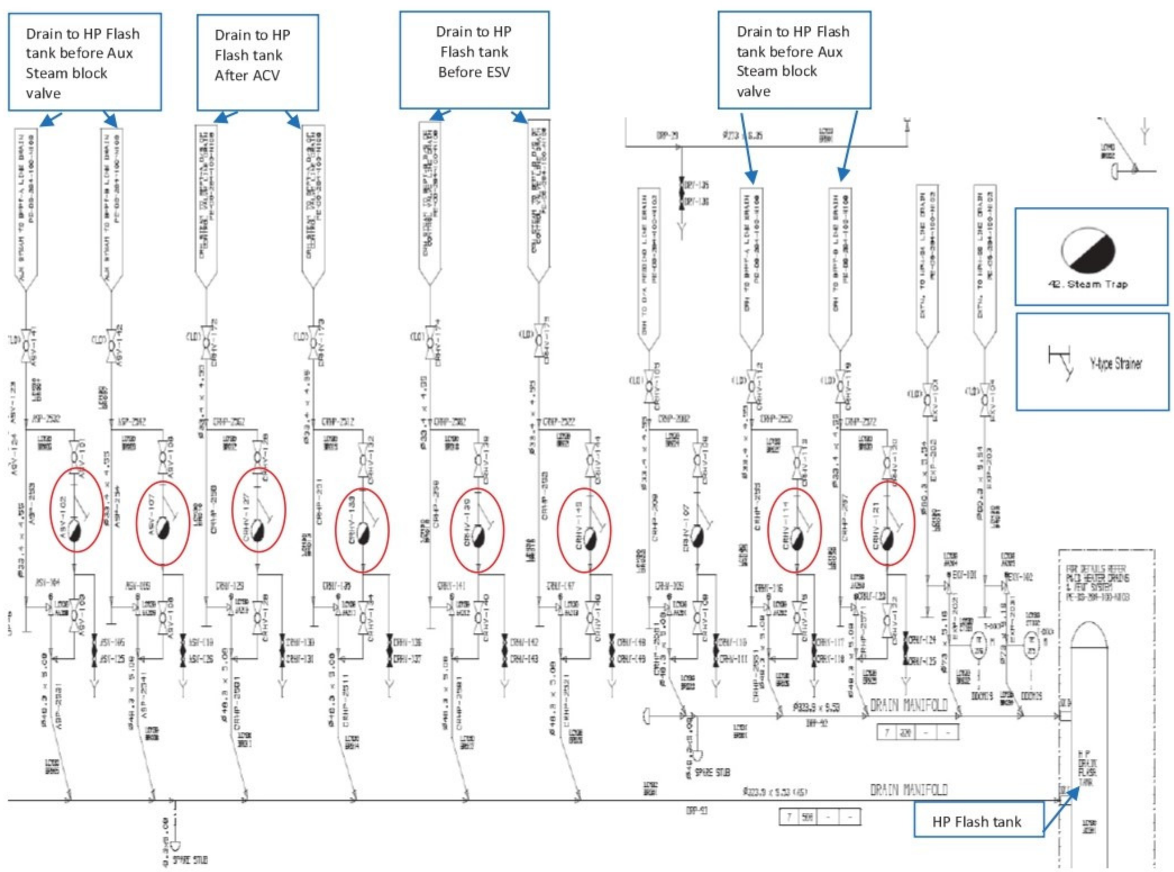


POINT	0	1	2	3	4	5	REMARKS
f%	0	14.5	44	55	65	100	%Feed back cam lift
f(mm)	0	2.001	6.072	7.59	8.97	13.8	Feed back cam lift(mm)
hv(mm)	0	10	28	32	43	75	Gov.valve lift (mm)
hz(mm)	0	17.33	48.53	55.47	74.53	130	Servomotor lift (mm)
Psec(atg)	1.5	1.94	2.83	3.15	3.46	4.5	Secondary oil pressure

ANNEXURE-2: Drain scheme of steam lines to TDBFP Drive Turbine (for Ref 500MW)



ANNEXURE-3: Details of Continuous drains / steam traps provided in drain scheme (for Ref 500MW)



<u>12.0 DISTRIBUTION LIST</u>

SIMHADRI INTRANET LMI PAGE